



Welcome to Orbital 26!



Outline

- General information
- The people factor
- Resources / Tools
- Level of achievements
- External Project
- Timeline of events
- Lift-off
- Mission control
- Milestones
- Splashdown

General Information

CP2106: Independent Software Development Project

- Teams of 2: Good teamwork and communication required.
- Workload: **140** in total or ~11 hours per week (per student)
- Compulsory events: Liftoff, 3 Milestones and Splashdown
- CS/CU grading

...you should have known all these by now since you have already applied and be accepted into the programme. ^_^

The People Factor: Facilitator

Dr. Zhao Jin

- zhaojin@nus.edu.sg
- Zhao Jin on MS Teams



The People Factor: Advisers

Advisers (**one per team**) are senior students with software development experience

- Serve 3 roles:



Evaluator



Project Manager



Cheerleader

The People Factor: Mentors

Mentors (*Artemis teams only*) are senior students, industry experts who want to pitch in and help guide student teams

- Provide more insights, resources and opportunities.

The People Factor: Tutors

Tutors are senior students, faculty members and industry experts with technical and project expertise

- Conduct workshops
- Provide technical and project support to teams

The People Factor: Peers

Other **teams** under the same adviser

- Meet up during group meetings, communicate via the Slack channel, etc.
- Discuss about ideas, share resources, give comments and feedback directly, etc.
- Give comments and feedback via Peer Evaluation

Resources / Tools

1. Email (IMPT)

- Announcements / Reminders

2. Canvas (IMPT)

- Announcements
- Discussions
- Lift-off / Mission Control

3. Skylab (IMPT)

- <https://nusskylab-v2-dev.comp.nus.edu.sg/>
- (Public) Project gallery
- (Internal) Submissions / Peer evaluation
- Use 'Forgot your password' with your NUSNET email address (e.g., e0...@u.nus.edu) to set your password and login accordingly.
- If you have entered your NUSNET id with a capital 'E' in your application form, please follow the same capitalization for login.

4. Git + Github (IMPT)

- Required tools for source code version control.
- To be covered in Mission Control #1

5. Mega list of software development resources

- <https://soc-n.us/orbital26-resources>

6. Orbital website

- <https://orbital.comp.nus.edu.sg/>
- General information

7. MS Teams (IMPT)

- Orbital 26 Team: Communications / activities for Artemis teams only
 - Relevant students will be added directly to the team.
- ~~— Orbital 26 (Interim) Team:~~
 - ~~— Discussions for all Orbital 26 students (via the QnA channel)~~
 - ~~— **Announcements all Orbital 26 students (via the announcements channel)**~~
 - ~~— Join the team using the code: **440htum**~~

8. Support for Entrepreneurship

- For teams who intend to commercialize their Orbital project (via a startup, App / Play Store, etc.)
- To-dos:
 - Think about the business side of your project early.
 - Indicate your interest at the end of Orbital
 - Obtain Intellectual Property from NUS and and apply for resources (e.g., funding, office space) from **SoC Furnace** and / or **NUS Enterprise** after Orbital

9. NOC (NUS Overseas Colleges)

- Slide Deck <https://soc-n.us/orbital26-liftoff-noc>
 - Password to be sent via email.
- Short sharing from NOC as part of the Live Q&A session on **12 May (Tue), 4pm**
- To-dos:
 - Attend the sharing
 - Get started with [entrepreneurship](#)
 - Apply for NOC later when the application opens.

10. Testimonials

- For programmes such as NOC / SEP / Internship / Scholarship
- Available to all Artemis teams, as well as Apollo 11 teams who have strong support from their advisers / have won awards during Splashdown.
- Provided on a just-in-time basis (not for self-keeping / future use)

11. Awards

- **Judges' Choice Awards:** Top Artemis teams (overall) as selected by the panel of judges
- **Honorable Mention Awards:** Most popular teams with as determined by the live voting during Splashdown
- Certificates and prizes will be given to the winners.

12. AI Tools

- **You are NOT allowed to use AI tools to generate codes for your project or content for your Milestone submissions.**
- However, you are allowed to use AI tools to
 - self-learn (e.g., to understand what is CI/CD)
 - integrate with them for your features (e.g., to call ChatGPT API get answers on behalf of the users)
 - generate assets for your project (e.g., portrait images for the characters in your game)

Levels of Achievement



Vostok¹



Project Gemini



Apollo 11



Artemis

Levels of Achievement

Features

(The requirements of a higher level implicitly include all requirements from the lower levels.)

- Vostok / Gemini
 - **1~2 / 3~5** feature of **sufficient complexity**¹ in total
 - Use of database (workload must be justified otherwise if there is no database)

- Apollo / Artemis
 - **6~8 / 9+** features of **sufficient complexity** in total

Levels of Achievement

Planning / Version Control

(The requirements of a higher level implicitly include all requirements from the lower levels.)

- Vostok / Gemini
 - GitHub repo + Basic version control (e.g., add / commit / push / pull)
- Apollo
 - Github issues with (monthly) milestones / labels / tags / assignee + Intermediate version control (branching, pull request)
- Artemis
 - 2-week sprint with objectives / allocation / tracking.
 - Github Projects + Code Review + CI/CD

Levels of Achievement

Design

(The requirements of a higher level implicitly include all requirements from the lower levels.)

- Vostok / Gemini
 - Use cases and features
 - Flow and architecture
- Apollo / Artemis
 - Design diagrams (drawn with tools)
 - Design principles + pattern
 - Design decisions

Levels of Achievement

Implementation

(The requirements of a higher level implicitly include all requirements from the lower levels.)

- Vostok / Gemini
 - Organization of files into folders
 - Code level comments
- Apollo
 - Coding standard
- Artemis
 - Code Review

Levels of Achievement

Testing

(The requirements of a higher level implicitly include all requirements from the lower levels.)

- Vostok / Gemini
 - System testing by the developers
- Apollo / Artemis
 - Multi-level (unit / integration / system) testing with automation + User testing
 - Proper test strategy (planning / test case design)

Levels of Achievement

Documentation

(The requirements of a higher level implicitly include all requirements from the lower levels.)

- Vostok / Gemini
 - Basic description of the system and the features
 - **10+/20+** pages of README

- Apollo / Artemis
 - Detailed description of features, quantification of complexity and justifications of design
 - SE evidence in every stage of the development process
 - **30+/50+** pages of README

Levels of Achievement

Quality of peer evaluation given

- Vostok / Gemini / Apollo / Artemis
 - Average feedback rating should be 2/3/4/4

Teamwork

- All levels
 - **Proper, consistent, and balanced contributions¹** to the project (esp. in terms of coding) from **both** members of the team with good communications and cooperation in **every** Milestone.

Levels of Achievement

- Most projects can be done to any level of achievement with the right amount of effort.
 - For past year projects, visit Skylab → Gallery (e.g., Select Cohort 2025).
- Discuss with your adviser as early as possible if you have any concerns.
- The LoA will be updated based on the Milestone Evaluations and finalized at Milestone 3.

Levels of Achievement

- For upgrading to **Artemis**, the team should show sufficient progress with respect to the requirements¹ in Milestone 1 & 2.
- Qualifying teams will be shortlisted and considered for Artemis in Milestone 3.
- No such requests will be entertained after Milestone 2.

Timeline of events

Mandatory events:

11-18 May - Liftoff

1 June - Milestone 1 - Ideation

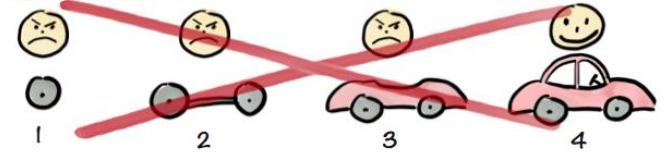
29 June - Milestone 2 - Prototyping

27 July - Milestone 3 - Extension

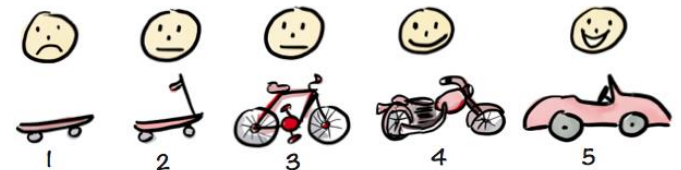
26 August - Splashdown - Refinement

Iterative Development

Not like this....



Like this!



Optional events:

16/23/30 May, 6 June, ... - Mission Control Workshops

Objective

- Get started with the project!

Activities

- (Compulsory) Programme overview
- (Optional) Live Q&A sessions
- (Compulsory) Meet-your-adviser
- (Compulsory) Poster & video submission
- (Compulsory for Artemis Teams) Mentor matching

Lift-off

Programme overview and live Q&A

- Read this set of slides carefully ^_^
- Post your questions in the Q&A channel in the Orbital 26 (Interim) Team (Team Code: **440htum**)
- Attend the live Q&A sessions for clarifications
 - **12 May (Tue), 4pm**
 - Via Zoom @ <https://soc-n.us/orbital26-liftoff-qna>
 - With a quick sharing from NOC
 - Recorded on a best effort basis
 - Make-up session on **15 May (Fri), 4pm** (same link).

Lift-off

Meet-your-adviser

- Have an online meeting / discussion with your adviser

Poster & Video submission

- An A4-size poster (<Project ID>.jpg or .png)
- A ~1-minute presentation video (<Project ID>.mp4)
 - Must be succinct and persuasive
 - Google “elevator pitch”, or “one minute madness” for more information

Example:

1999.jpg (poster) and
1999.mp4 (video).

See the next few slides for instructions
about how to find your Project ID on Skylab.

Orbital 25 submissions: <https://soc-n.us/orbital25-liftoff-playlist>

Poster & Video submission

- Focus on your project idea, basic information, preliminary design and plan for now.
- Also required for future [Milestone submissions](#) (w/ harder requirements).
- General tips / tools from CFG: <https://soc-n.us/orbital26-liftoff-poster-video>
- Official Workshop in Mission Control #3

Poster & Video submission

- Deadline: **18 May (Mon), 2pm**
- Upload the files to your Google Drive, and submit the links to **Skylab**
 - The files should be **named correctly**
 - The links should be **Google Drive File Links** and **publicly accessible**

**Teams who do not follow the instructions closely
are subject to removal**

To find out your project ID on Skylab

- After logging into Skylab, click on **Team Profile** in navigation bar in the upper right corner.

STAFF

TEAM PROFILE

PROFILE

SIGN OUT

- The project ID (a 4-digit number) will be shown on the profile page together with other information of your team.

Lift-off

(Artemis Teams only) Mentor matching

- Complete the technical proof of concept by **13 May (Wed)**
- Prepare the project materials and share them with the mentors via the mentorship matching channel in MS Teams by **14 May (Thu), 2pm**

Lift-off

(Artemis Teams only) Mentor matching

- Communicate with the mentors to reach an agreement
- Contact Jin to confirm the allocation by **25 May (Mon), 2pm.**

Mission Control

Objective

- Get some technical help and project advice!

Activities

- Technical help sessions
- Project consultations

(More details will be released later...)

Mission Control

Workshops

- Crash courses on specific technologies (e.g., Git, React, Unity) from the tutors
- Sign-up and proper preparation required.
- More details will be released later...

- Via Zoom and recorded on a best-effort basis
- Release of materials in the following week

Mission Control

Project Consultations

- 20 minutes per session
 - Get comments and advice specific to your project from a tutor / adviser
 - Sign-up and proper preparation required.
 - More details will be released later...
-
- Via Zoom but not recorded

Mission Control

Mission Control #1 (16 May):

- Web App Development: React
- Mobile App Development: Flutter, Part 1
- Database: Relational Databases
- General Tools: Git / GitHub

Info doc: <https://soc-n.us/orbital26-mc1-info>

Mission Control

Mission Control #2 (23 May):

- Game Development: Unity
- Web App Development: NextJS
- Mobile App Development: Flutter, Part 2
- Database: NoSQL
- AI Features in Apps: Part 1

Mission Control

Mission Control #3 (30 May):

- Machine Learning: Part 1
- Game Development: SE in Game Dev
- Software Engineering: Part 1
- Deployment: Part 1
- AI Features in Apps: Part 2

Mission Control #3 (3 Jun):

- Project Poster and Video

Mission Control

Mission Control #4 (6 Jun):

- Machine Learning: Part 2
- Software Engineering: Part 2
- Deployment: Part 2

Mission Control #5 (early-Aug):

- Orbital Resume Workshop

Milestones

Objective

- Give a progress update and get feedback.

Activities

- Project work
- Milestone submission
- Milestone evaluation

Milestones

Project work by 1 Jun - Milestone 1 (Ideation)

- Formulate your project idea clearly
- Identify the features for your system
- Design your system
- Create a development plan
- Pick up the necessary technologies
- Build a **technical proof of concept** (e.g., an integrated frontend+backend with the login/register feature)
- Document your system

Milestones

Project work by 29 June - Milestone 2 (Prototype)

- Implement the prototype of your system, which should contain the most essential features
- Perform system testing
- Document your system

Milestones

Project work by 27 July - Milestone 3 (Extensions)

- Extend your system by adding more useful features
- Perform system / user testing
- Document your system

Project work by 26 August - Splashdown (Refinement)

- Polish up your system and fix outstanding issues
- Make your final submission required

Milestones

Apollo 11 teams are expected to complete **more work of higher quality within the same timeline.**

Artemis teams are expected to complete **more work of higher quality roughly one Milestone ahead** (e.g., completion of technical proof-of-concept within Liftoff).

Milestone Submission

Milestone X



Submit **the links of the deliverables** to **Skylab** by the following deadlines

- Milestone 1: 1 June, 2pm (SGT)
- Milestone 2: 29 June, 2pm (SGT)
- Milestone 3: 27 July, 2pm (SGT)

Milestone Submission

Milestone X



The **deliverables** include:

- README
- Project Log
- Project Poster
- Project Video

The deliverables for a Milestone should include all contents from the previous Milestones (possibly in a condensed form) as well.

What should be included in README?

- Your targeted level of achievement, as well as
 - Milestone 1 (Ideation): include problem motivation, proposed core features / user stories, design and plan
 - Milestone 2 (Prototype): include core features developed, problems encountered
 - Milestone 3 (Extension): include bugs squashed, edge features developed, problems encountered, and user testing

... and any other info that justify your LoA!

What should be included in Project Log?

- What were the tasks performed by each member?
- When were the task performed?
- How many hours were spent in completing each task?

The total number of hours spent per student should be at least 140 hours at the end of the programme, and

both members are expected to make proper, consistent, and balanced contributions to the project.

A Very Bad Example

- We are planning to build a website for selling secondhand textbooks.
- Project Log
 - Team Meeting: 40 hours.

Good Examples

<https://soc-n.us/orbital-ms-sample-1>

Note: This is the bare minimum for Project Gemini.

<https://soc-n.us/orbital-ms-sample-2>

Note: This is sufficient for Artemis.

Excellent Examples

<https://soc-n.us/orbital-ms-sample-3>

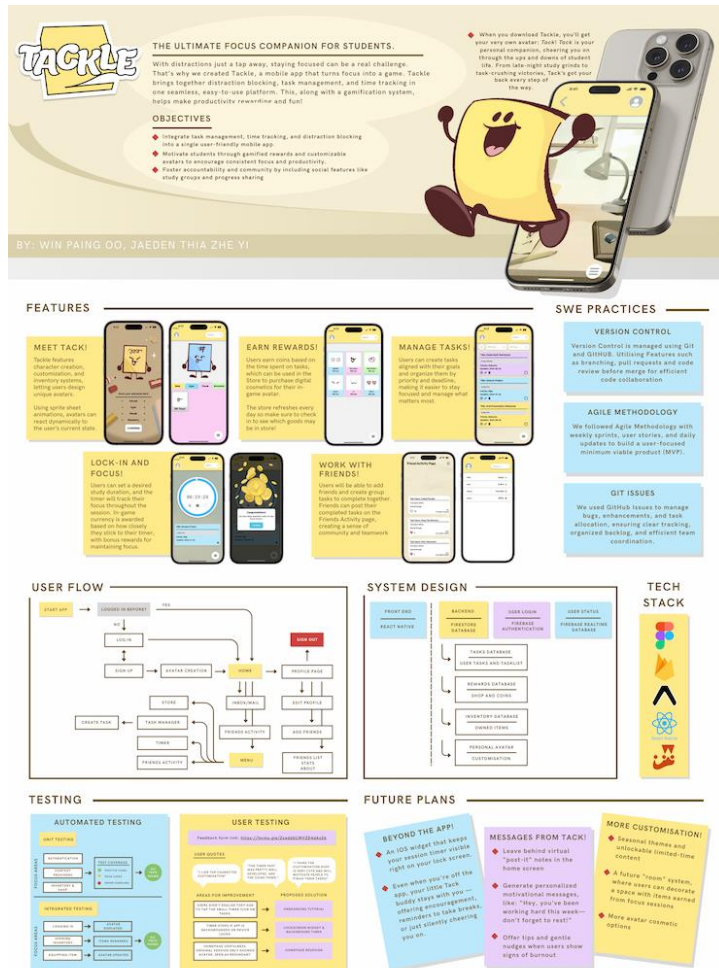
<https://soc-n.us/orbital-ms-sample-4>

Note: This is written by the top **Artemis** Teams from Orbital 22 for **Milestone 3**.

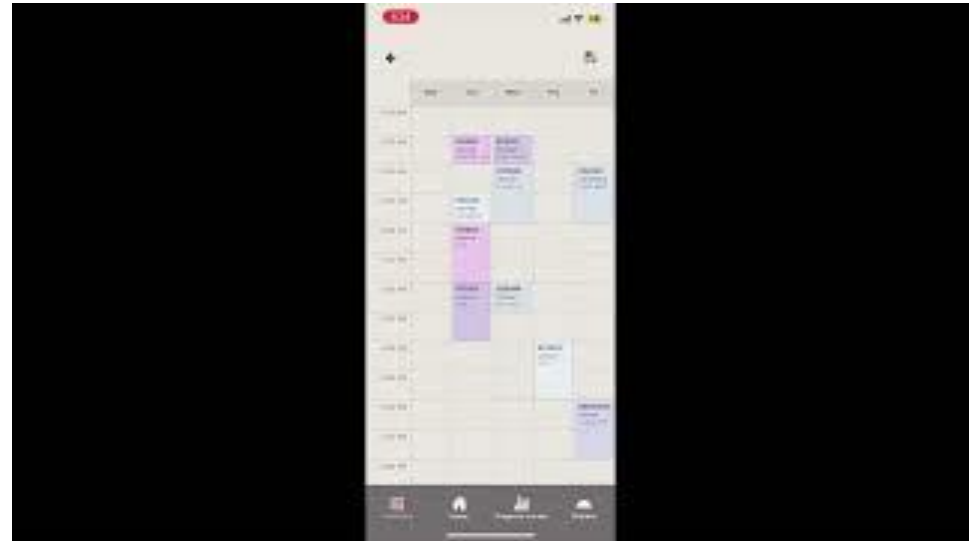
Project Poster & Video

- Milestone 1: Showcase your formulated project idea and technical proof of concept
 - Similar to Liftoff submissions but more details required.
- Milestone 2 & 3: Showcase your prototype / complete system
 - Poster: **A1 size**, multi-panel / segment
 - Video: ~5 minutes

An Excellent Example



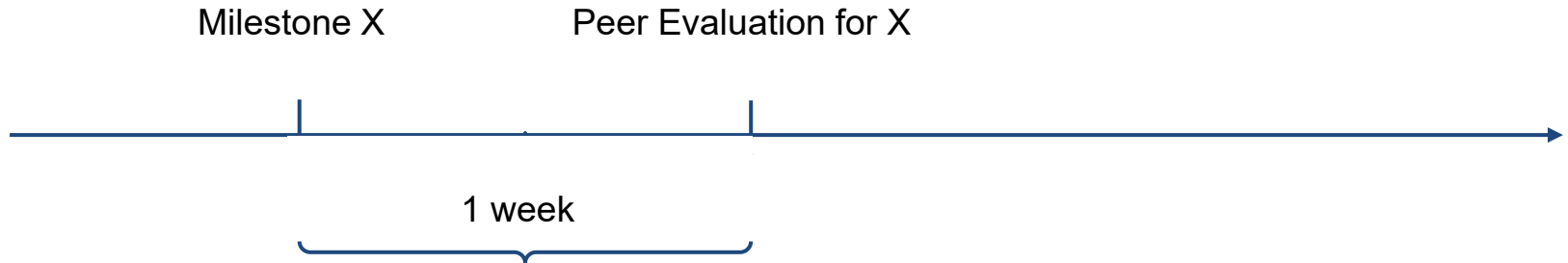
Winners of Best Project Poster / Video Award from Orbital 25



What will be asked in the evaluation form?

- Is the problem well defined? Is it clear what need the system is trying to satisfy? Is there a real need to solve the problem?
- Are the features of the system clearly specified?
- Is there a draft plan for next development cycle?
- **Sample Evaluation Forms**
 - Milestone 1: <https://soc-n.us/orbital-ms-eval-1>
 - Milestone 2: <https://soc-n.us/orbital-ms-eval-2>
 - Milestone 3: <https://soc-n.us/orbital-ms-eval-3>

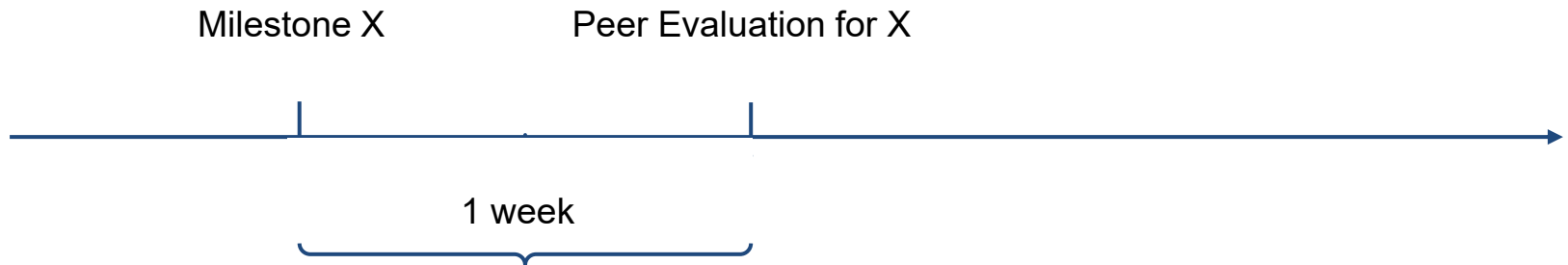
Peer / Adviser Evaluation



Teams also evaluate their peers (3 other teams)

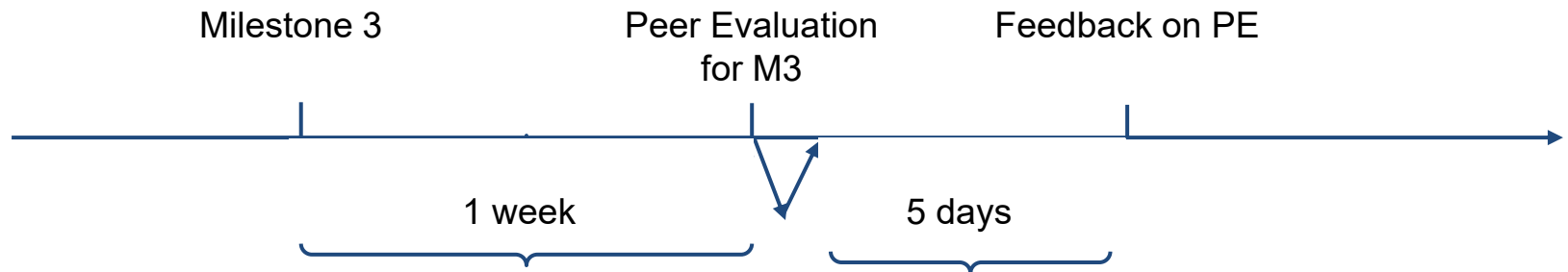
- Fill out the evaluation form on Skylab based on the materials presented by each team.
- Helpful and critical constructive comments required for higher levels of achievement.

Negative Feedback



- Constructive (negative) feedback is also part of the form, and is source-anonymized for the receiving team.
 - Advisers and other staff can see identity of anonymous contributors.

Feedback on Peer Evaluation



- Final milestone has additional feedback cycle; i.e., evaluate the peer evaluators.
- Obtaining good peer feedback scores is also part of the higher levels of achievement.
- Most teams will evaluate the same peer teams throughout Orbital.

Splashdown

Objective

- Showcase your project, win awards and get ready for the next stage!

Activities

- Poster presentation
- Invited talks
- Award ceremony

FAQs

- Level of achievement / mentorship
 - Can we upgrade to Artemis later without any mentor guiding us?
 - Yes.
 - For artemis teams, is it one team to one mentor or can one mentor have multiple teams?
 - Most mentors would only take one team so please do make effort in persuading them.
- Can an Artemis team with mentorship drop to Apollo 11 before milestone 3?
 - No. An Artemis team with mentorship is required to persist through and be evaluated at Artemis level.

- Level of achievement / mentorship
 - What is considered evidence of software engineering for Apollo 11 and Artemis?
 - Software engineering is the systematic application of engineering approaches to the development of software. Please check out the [CS2103 website](#) for the core concepts and recommended practices. You should provide details in your Milestone submission showing that you understand the concepts and have applied the recommended practices.
- Will advisers recommend the suitable level of achievement for our project?
 - Yes. Please discuss with your adviser proactively.

- Project Work
 - Can we change our project details? (Updated on 11 May)
 - Yes. The details (e.g., features, level of achievement, tech stack) can be changed; However, change of topic / project type is generally not allowed.
 - However, the more changes you are making the more time you need to work out the details / learn the required technologies. You might end up having difficulties in meeting the stated deadlines (which will not be extended for such reasons).
 - Nevertheless, if your concern is that your existing idea is too difficult, please discuss with your adviser to simplify it first.

FAQs

- Project Work
 - How do I get started with the project work?
 - You can start by figuring out as many details about your project as possible. For example: What are the key features? what is the system design? What is the tech stack? What good software engineering practices your team will be following?

FAQs

- Project Work
 - How do I identify the most suitable tech stack for my project?
 - For most Orbital projects, any tech stack should be sufficient. But do take note that some might be more difficult to learn.
 - Can I use resources (e.g., external APIs, real-life data, game assets) that are not created by my team?
 - Yes. But **you must give credits properly** to the resources that you have used. Such resources won't be counted as the work you have done in Orbital (but the effort of integration may still be counted if it is non-trivial).

- Project Work
 - What happen if my project require permissions to use such resources?
 - It is non-trivial to obtain relevant permissions / pay for the license fee so you might need to choose something publicly available or have some backup plans.
 - Here are a few possible alternatives for your consideration.
 1. Check if there are unofficial wrappers for the APIs (and do not spam requests to avoid being caught).(Answer continues on the next slide)

FAQs

2. Design your own mock APIs + Database (with dummy data) so that your system uses those instead of communicating with the actual APIs. The dummy data can be manually created / scraped from some webpages. (If you are aiming for higher levels, you can even design your system in a way such that it is easy to plugin any APIs as needed.)

3. Shift the focus of your project / redesign your feature. This is part of the flexibility of Orbital. So please don't be afraid to do so. For example, if you cannot obtain real time data for bus arrival, you can make use of historical data / the schedule and ask your user for input to update the estimates.

FAQs

- Project Work
 - Will the costs of the project be covered by Orbital?
 - No. You will need to settle all the costs for your project by yourself.
 - Note that you are already taking Orbital without having to pay the special term fees. Therefore, please make use of the money saved for your project.

- Project Log for Milestone submission
 - Do we have to specify how many hours each member has spent on a particular task?
 - Yes. The workload requirement of 140 hours is for each student in the team.
- Can anything related to the project be recorded in the log?
 - Yes. Anything related to the project / Orbital can be recorded in the log. But please make sure that a substantial portion of the time is on project work (e.g., it does not make sense to claim that you have self-studied for 135 hours but only done 5 hours of project work).

All the best!
